

Research Proposal

Evaluation of Low-Cost MEMS Accelerometers for Structural Vibration Analysis and Educational Applications

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June 13, 2026

Abstract

Mechanical vibration analysis is a critical topic in engineering fields such as machine design, rotating machinery diagnostics, aerospace structures, manufacturing systems, and condition monitoring. Professional vibration measurement systems often require specialized accelerometers and data acquisition hardware that may be prohibitively expensive for educational institutions and student projects.

This research proposes the evaluation of the MPU6050 MEMS accelerometer as a low-cost alternative for vibration analysis applications. A structural vibration analyzer prototype has already been developed using three MPU6050 sensors, an ESP32 microcontroller, and custom Python-based signal processing software. The primary objective of this research is to experimentally determine the limitations, accuracy, and practical capabilities of the MPU6050 when compared to analytical predictions and, when available, professional vibration measurement equipment.

The project also seeks to investigate the potential use of low-cost vibration analyzers as educational tools in Mechanical Vibrations courses. The expected outcome is a validated assessment of the strengths and weaknesses of MEMS-based vibration measurement systems and recommendations regarding their suitability for engineering education and introductory vibration studies.

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1 Introduction

Mechanical vibration analysis is commonly used to identify natural frequencies, resonance conditions, structural response characteristics, and machinery faults. Traditionally, vibration studies rely on laboratory-grade instrumentation systems, including piezoelectric accelerometers and dedicated data acquisition hardware.

Recent advances in Micro-Electro-Mechanical Systems (MEMS) have resulted in the availability of inexpensive accelerometers such as the MPU6050. Although these sensors are widely used in robotics, drones, and motion sensing applications, their suitability for engineering vibration analysis remains an area of interest.

This project builds upon a structural vibration analyzer prototype previously developed by the author. The system utilizes multiple MPU6050 accelerometers, an ESP32 microcontroller, and Python-based Fast Fourier Transform (FFT) analysis tools to measure structural vibrations.

The proposed research seeks to quantify the capabilities and limitations of this low-cost system and determine whether such a platform can serve as a viable educational tool for undergraduate engineering courses.

2 Research Problem

Commercial vibration analysis systems are often costly and inaccessible to many educational institutions and students.

Although low-cost MEMS accelerometers offer an attractive alternative, questions remain regarding:

- Measurement accuracy
- Frequency range limitations
- Noise characteristics
- Sensitivity to low-amplitude vibrations
- Ability to identify natural frequencies
- Repeatability and reliability

Currently, limited information is available regarding the practical use of MPU6050 sensors for engineering vibration analysis applications.

3 Research Objectives

3.1 Primary Objective

Determine the limitations and practical capabilities of the MPU6050 accelerometer for structural vibration analysis.

3.2 Secondary Objectives

1. Develop controlled vibration experiments using structures with analytically predictable behavior.
2. Compare measured natural frequencies against theoretical calculations.
3. Evaluate measurement repeatability and consistency.
4. Quantify sensor noise and frequency resolution limitations.
5. Investigate the effects of sensor placement and sensor mass loading.
6. Compare results with professional instrumentation when available.
7. Develop laboratory procedures suitable for undergraduate engineering education.
8. Assess the feasibility of implementing low-cost vibration analyzers in Mechanical Vibrations and Experimental Methods courses.

4 Research Questions

1. Can an MPU6050 accurately identify structural natural frequencies?
2. What frequency range can be reliably measured?
3. What is the minimum vibration amplitude that can be detected?
4. How repeatable are measurements obtained from MPU6050 sensors?
5. How closely do measured frequencies match analytical predictions?
6. Can low-cost vibration analyzers provide educational value comparable to higher-cost laboratory equipment?

5 Prototype Developed

A functional structural vibration analyzer has already been developed.

5.1 Hardware

- ESP32 Development Board
- PCA9548A I2C Multiplexer
- Three MPU6050 Accelerometers
- USB Serial Communication Interface
- Custom Sensor Mounting Fixtures
- Desktop Computer for Data Processing

5.2 Software

- ESP32 Firmware (C++)
- Python Data Logger
- Python FFT Analysis Software
- NumPy
- SciPy
- Pandas
- Matplotlib

5.3 Current Capabilities

The current prototype can:

- Acquire vibration data from up to three accelerometers simultaneously.
- Record synchronized acceleration measurements.
- Export vibration data to CSV format.
- Perform Fast Fourier Transform analysis.
- Generate frequency-domain plots.
- Compare vibration responses at multiple locations.

6 Research Methodology

6.1 Phase I: Analytical Validation

Simple structures with known analytical solutions will be used.

Examples include:

- Cantilever beams
- Mass-Spring

Theoretical natural frequencies will be calculated.

Measured frequencies will then be compared with analytical predictions.

6.2 Phase II: Experimental Validation

Controlled vibration tests will be conducted.

Methods may include:

- Impact hammer testing
- Free vibration decay
- Speaker-induced vibration

Data will be collected using the developed analyzer.

6.3 Phase III: Repeatability Analysis

Repeated experiments will be performed to evaluate:

- Frequency measurement consistency
- Amplitude consistency
- FFT peak stability

6.4 Phase IV: Comparative Validation

When professional equipment becomes available, direct comparisons will be performed.

Potential comparison systems include:

- National Instruments DAQ systems
- LabVIEW-based vibration systems
- Industrial accelerometers
- Piezoelectric vibration sensors

7 Educational Applications

One of the motivations for this research is the potential use of low-cost instrumentation within engineering education.

Possible laboratory activities include:

1. Determination of natural frequencies.
2. Experimental modal analysis demonstrations.
3. FFT signal analysis exercises.
4. Resonance investigations.

5. Vibration isolation experiments.
6. Beam vibration laboratories.
7. Data acquisition and signal processing instruction.

The resulting platform could provide students with hands-on experience at a fraction of the cost of traditional laboratory equipment.

8 Future Improvements

Several improvements are planned to expand the capabilities of the current prototype.

8.1 Hardware Improvements

- MicroSD data logging
- OLED user interface
- Rotary encoder control system
- Battery-powered operation
- Wireless communication
- Improved sensor mounting systems
- Alternative MEMS accelerometers
- Higher-performance accelerometers

8.2 Software Improvements

- Embedded FFT processing on ESP32
- More advanced microcontrollers
- Automated modal identification
- Improved filtering algorithms
- Data synchronization improvements
- Advanced transmissibility calculations
- Frequency response function estimation

8.3 Validation Improvements

- Comparison with laboratory-grade accelerometers
- Comparison with National Instruments systems
- Calibration procedures
- Uncertainty analysis
- Statistical error quantification

9 Expected Outcomes

The project is expected to produce:

1. A quantified assessment of MPU6050 vibration measurement capabilities.
2. Identification of measurement limitations.
3. Validation of the developed structural vibration analyzer.
4. Educational laboratory procedures.
5. Recommendations regarding the use of low-cost vibration instrumentation in engineering education.
6. A foundation for future research involving advanced accelerometer technologies.

10 Expected Deliverables

- Final research report
- Experimental test procedures
- Validation datasets
- FFT analysis software
- Structural vibration analyzer hardware
- Educational laboratory modules
- Recommendations for curriculum implementation

11 Conclusion

This research seeks to bridge the gap between low-cost sensor technology and engineering vibration analysis by evaluating the MPU6050 accelerometer in a controlled experimental environment. Through analytical validation, experimental testing, and future comparison with professional instrumentation, the project aims to establish the capabilities and limitations of MEMS-based vibration analyzers.

The results have the potential to contribute both to engineering education and to the development of affordable vibration analysis tools for academic institutions, student projects, and introductory research applications.